

Date: 2026-04-28
Request ID: 100048419

Hotplate		
Element(s)	Pt	
SOP#		
Cocktail	15 mL HCl, 5mL HNO3, 10 mL H2SO4	
sample(s)	weight (g)	volume (mL)
200127553_1	0.204	100
200127553_2	0.2028	100

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Pt ICP analysis

sample	Dilution	conc. [mg/L]	Pt_ppm	%Pt
200127553_1	10/8	2.291	1403.80	0.14
200127553_2	10/8	2.83	1744.33	0.17

pt result: 1574.06 ppm

pt result: 0.16 %

pt oxide factor:

pt oxide result: 0.00 %

pt lot: *Manufacturer: Assurance, 28-2PTY exp: 2027-02-28*

Note(s): *Microwave digestion was not successful, samples exploded out of the PTFE vials.*

A = crucible

B = crucible + sample

C = crucible + sample after drying

$$\%LOI = ([B-A] - [C-A]) / (B-A) * 100\%$$

$$ppm\ M+ = [conc.][volume][dilution] / [weight]$$

$$\%M+ = [conc.][volume][dilution] / ([weight] * 10,000)$$

$$\%MO = [oxide\ factor] * \%M+$$

$$ppm\ M+ \ [dried] = ppm\ M+ / ([1 - (\%LOI) / 100])$$

$$\%M+ \ [dried] = \%M+ / ([1 - (\%LOI) / 100])$$

$$\%MO \ [dried] = \%MO / ([1 - (\%LOI) / 100])$$

$$\%Cr(VI) = (1.733[mL\ FAS][N\ FAS]) / ([weight])$$

$$\%Cr_2O_3 = 1.462 * [\%Cr(VI)]$$

$$\%Cr(III) = total_Cr - Cr(VI)$$